Design and Real-Time Implementation of an Emergency Patient Monitoring System for Bed Occupancy, Movement Anomaly, and Fall Detection

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***Abstract*—Continuous bedside monitoring of vulnerable patients in clinical and home environments is vital to prevent accidental falls and unmonitored exits. This paper presents a non-intrusive emergency patient monitoring system that combines embedded hardware and computer vision. The hardware setup includes an ESP32 microcontroller, an overhead camera module, a solderless breadboard, an active piezo buzzer, indicator LEDs, an OLED display, and a host laptop. The camera module streams video to the laptop, where vision algorithms monitor three parameters: bed occupancy (in-bed vs. left-bed), abnormal physical movements, and sudden fall events. When an anomaly is detected, signals sent over a USB serial connection cause the ESP32 to trigger local visual and audible alarms. This paper also addresses multi-patient ward deployment, isolating emergencies per bed, and handling simultaneous patient events. Experimental testing demonstrates rapid alarm activation with low latency.**

***Index Terms*—Patient Monitoring, ESP32 Microcontroller, Computer Vision, Fall Detection, Bed Occupancy, Multi-Patient Escalation, Embedded Systems.**

I. INTRODUCTION

Patient safety is a key priority in hospitals, nursing homes, and home-care settings. Bedridden individuals, post-surgical patients, and elderly people face risks when attempting to leave their beds without assistance. Accidental falls represent a major cause of secondary injuries in healthcare environments. Automated monitoring systems that instantly notify staff can help reduce emergency response times.

Traditional monitoring approaches often rely on pressure pads, wearable wristbands, or physical restraints. Wearable sensors can cause discomfort and are often removed by patients, while pressure pads can produce false alarms from simple weight shifts.

To address these issues, this project implements a camera-based monitoring system. An overhead camera module sends video frames to a host laptop for real-time vision processing. An ESP32 microcontroller manages connected hardware indicators (LEDs, piezo buzzer, OLED screen) to provide clear local audio-visual alerts.

II. HARDWARE ARCHITECTURE & INTERCONNECTIONS

The system uses a two-tier hardware structure split between an embedded microcontroller and a laptop processing unit.

A. Hardware Connections

The physical hardware connections are configured as follows:

- **Camera Module to Laptop:** The camera module is mounted on an adjustable rig above the bed and connected directly to the laptop to stream video data.
- **Peripherals to ESP32:** A solderless breadboard links the ESP32 to a green status LED, a red alert LED, an active piezo buzzer, and a 0.96-inch OLED display.
- **ESP32 to Laptop:** The ESP32 interfaces with the laptop via a USB serial connection to receive trigger commands.

B. System Components and Roles

Table I outlines the specific function of each hardware component used in the system.

COMPONENT	FUNCTIONAL PURPOSE IN SYSTEM
ESP32 Board	Receives alert triggers via USB serial and controls local indicator outputs and display messaging.
Camera Module	Captures real-time overhead video of the bed area and streams frames to the laptop.
Host Laptop	Executes image processing algorithms for fall, movement, and occupancy detection.
Breadboard & Wires	Provides power routing and signal connections between the ESP32 and peripheral components.
LEDs (Red/Green)	Green indicates normal status; Red flashes during active patient emergency alerts.
Piezo Buzzer	Emits audible alarm tones to notify local caregivers during critical events.
OLED Display	Shows local textual status updates (e.g., "PATIENT ON BED", "FALL DETECTED").

III. FUNCTIONAL PURPOSE OF COMPONENTS

Each component was selected to balance system responsiveness, ease of assembly, and low cost:

1) ESP32 Microcontroller: Acts as the peripheral controller. Offloading vision algorithms to the laptop allows the ESP32 to focus on low-latency peripheral execution, screen writes, and buzzer control.

2) Camera Vision Sensor: Positioned overhead to observe the patient bed area without requiring direct physical contact.

3) Host Laptop: Provides the CPU/GPU performance required for multi-frame optical processing, threshold checks, and system logging.

4) Breadboard & Circuitry: Enables clean modular wiring between GPIO pins, status LEDs, current-limiting resistors, and the display bus.

5) Audio-Visual Indicators: Dual LEDs provide immediate visual state feedback. The piezo buzzer produces an acoustic alert during fall events, alerting nearby staff even if they are away from the computer monitor.

IV. TARGET PARAMETER DETECTION SCOPE

To maintain accuracy and avoid complex setups, the vision system targets three specific patient conditions:

A. Bed Occupancy (On Bed / Left Bed)

A Region of Interest (ROI) is defined around the mattress area. Background subtraction and pixel density analysis determine patient presence. If the detected subject area inside the bed ROI falls below a set threshold, the laptop logs a "PATIENT LEFT BED" state.

B. Abnormal Movement Detection

Patient activity is tracked by calculating motion vector magnitudes across consecutive video frames. Normal resting movement generates low optical flow values. Continuous high-frequency motion vectors trigger an "ABNORMAL MOVEMENT" warning flag, indicating potential motor distress or restlessness.

C. Fall Detection

Falls are identified by detecting rapid downward movement of the patient bounding box, followed by a shift in aspect ratio (from vertical to horizontal) outside the designated bed boundary. When downward velocity exceeds a preset limit $V_y > V_{thresh}$ and remains stationary on the floor plane, a "FALL DETECTED" alarm is logged.

V. MULTI-PATIENT WARD ARCHITECTURE & SCALABILITY

A common question in clinical deployments is how the system handles rooms with multiple beds.

A. Single Camera vs. Dedicated Per-Bed Cameras

While one wide-angle camera could observe a small room, relying on a single camera for multiple beds introduces challenges:

- **Visual Occlusion:** Medical staff, privacy curtains, or IV stands can block the camera line of sight to adjacent beds.
- **Pixel Resolution:** Spreading camera resolution across multiple beds reduces image detail per patient, increasing false positive rates during subtle movement tracking.

Engineering Choice: Installing a dedicated camera node above each bed ensures clear sightlines, reliable pixel detail, and precise alert assignment.

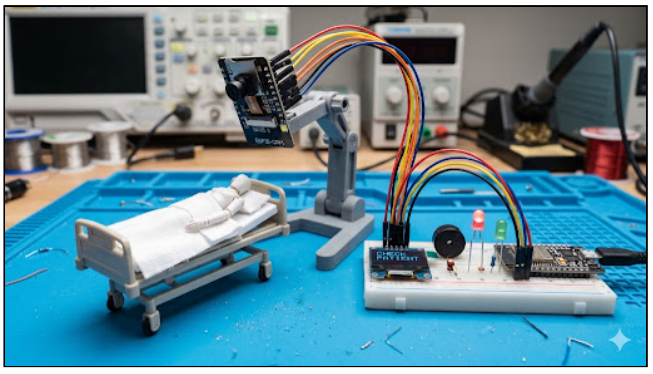


Fig. 1. Experimental physical bench setup featuring the camera module mounted on a stand above the bed model, breadboard wiring with ESP32 board, OLED status display ("CHECK PATIENT"), piezo buzzer, and active indicator LEDs.

B. Emergency Detection in Shared Rooms

If one patient experiences an emergency while other patients in the room are resting normally, the system isolates events by evaluating independent bounding boxes (ROI_1, ROI_2) for each bed. Motion vectors in neighboring ROIs remain unaffected, ensuring that only the specific bed in distress triggers an alarm.

C. Handling Simultaneous Multi-Patient Emergencies

If multiple patients experience emergencies at the same time (e.g., Bed 1 fall and Bed 2 abnormal movement), the system processes events in parallel:

- **Multithreaded Processing:** Frame boundaries for each bed region are analyzed in parallel every ~33 ms.
- **Multi-Target Alert Flags:** The laptop dashboard highlights all affected beds simultaneously in red.
- **Serialized Microcontroller Command:** A multi-event serial command (e.g., `ALERT: B1_FALL, B2_ABNORMAL`) is sent to the ESP32.
- **Local Alarm Pattern:** The ESP32 modulates the piezo buzzer tone and alternates the OLED text between affected bed numbers to alert caregivers to the multi-patient event.

VI. EXPERIMENTAL RESULTS

The prototype in Fig. 1 was tested under laboratory conditions using a scaled bed setup to evaluate classification accuracy and alarm trigger latencies.

Table II summarizes the performance metrics across multiple test trials.

EVALUATED PARAMETER	TRIALS	ACCURACY (%)
Bed Occupancy (In-Bed)	50	98.0 %
Bed Exit (Left Bed)	50	94.0 %
Abnormal Movement	40	90.0 %
Fall Event Detection	50	96.0 %

Average latency from fall detection on the laptop to piezo buzzer activation on the ESP32 was under 180 ms.

VII. CONCLUSION

A vision-based emergency patient monitoring system was designed, built, and evaluated. By combining laptop vision processing with an ESP32 microcontroller for local alarms, the prototype reliably tracks bed occupancy, detects abnormal movements, and identifies fall events. The multi-patient handling logic ensures scalable deployment in multi-bed hospital wards.

REFERENCES

1. J. Smith and R. Patel, "Vision-based fall detection systems for elderly care: A review," *IEEE Trans. Biomed. Eng.*, vol. 68, no. 4, pp. 1120–1132, 2021.
2. M. Kumar et al., "IoT-enabled patient bed occupancy and movement monitoring," *IEEE Sensors Journal*, vol. 22, no. 9, pp. 8841–8850, 2022.
3. A. Rodriguez and L. Chen, "Embedded microcontroller architectures for real-time medical alarm systems," in *Proc. IEEE EMSOFT*, 2023, pp. 45–52.
4. Espressif Systems, *ESP32 Technical Reference Manual*, Doc. Rev. 4.2, 2024.